

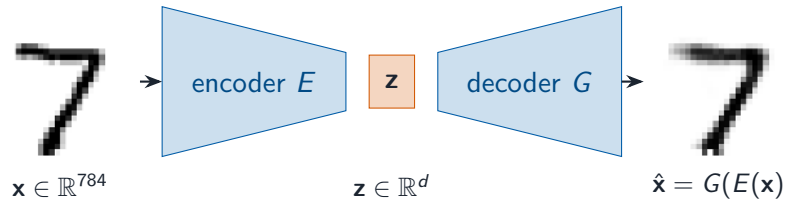

Lecture 6: Generative Models
Variational and Conditional Autoencoders
Generated Spins and the Hamiltonian

# Deep Learning in Classical and Quantum Physics

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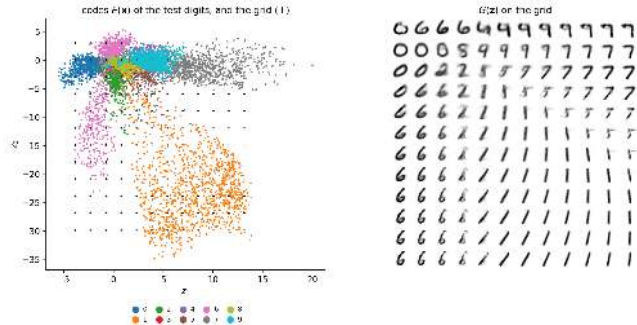
# Recap: the Autoencoder and its Latent Space




- trained on images alone, $\mathcal{L} = \frac{1}{n}\sum_i \frac{1}{784}\|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$; the bottleneck $d \ll 784$ is a learned, lossy compression (Lecture 5, Lab 017)
- similar images get nearby latent vectors $\mathbf{z}$; the decoder turns *any* $\mathbf{z}$ into an image

*Fed with latent vectors that no training image was mapped to, does the decoder produce new digits?*

# Decoding the Latent Plane of an Autoencoder




Autoencoder with $d = 2$. Left: $\mathbf{z} = E(\mathbf{x})$ for the 10 000 test digits; right: $G(\mathbf{z})$ on a $12 \times 12$ grid covering the codes. Coordinates of very different spread, $\mathrm{std}(z_1, z_2) = (3.8, 7.5)$; the 1s fill most of the plane, the other digits crowd together; between clusters: mixtures. **No distribution tells us where, and how often, to sample.**

# Random Latent Vectors Give No Digits




Autoencoder with $d = 16$, decoder applied to random $\mathbf{z}$. Top: $\mathbf{z} \sim \mathcal{N}(0, I)$; bottom: $z_k \sim \mathcal{N}(m_k, s_k^2)$, with $m_k, s_k$ the mean and standard deviation of $E(\mathbf{x})_k$ over the training set.

- the codes lie on a thin, curved set in $\mathbb{R}^{16}$; a random point almost never falls on it, even with the right mean and spread per coordinate
- a CNN classifier (Section 3) is confident, $\max_c p(c \mid \mathbf{x}) > 0.9$, for 37% of such images, against 96% of real test digits

**To generate we need a latent space with a known distribution.**

# 1

# Learning a Probability Distribution



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# Generative Models

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **samples** from an unknown probability distribution $p(\mathbf{x})$: handwritten digits, or spin configurations at temperature $T$, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$.

A **generative model**

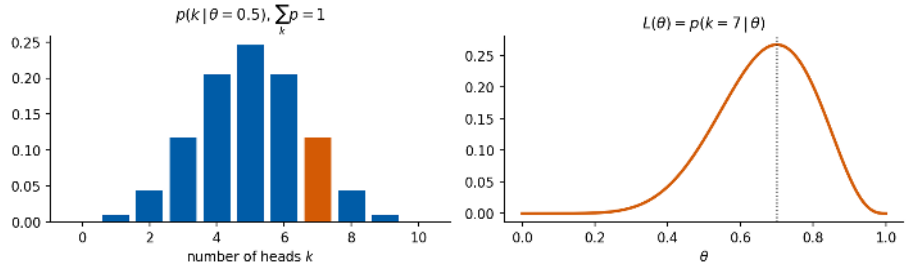
- learns an approximation $p_\theta(\mathbf{x}) \approx p(\mathbf{x})$ from the samples ($\theta$: its parameters)
- draws **new samples** from it: instances that are not in the data set, but could have been

In science: new molecules and materials, ensembles of weather forecasts, fast surrogates of expensive simulations, and Monte Carlo samplers for lattice models.

**Two steps: fit a distribution to data (maximum likelihood), and make it easy to sample (latent variables).**

# Probability and Likelihood

A model $p(x \mid \theta)$ is read in two ways: **probability** ($\theta$ fixed, $x$ varies, $\sum_x p = 1$) and **likelihood** (data fixed, $\theta$ varies, $L(\theta) = p(x_{\text{obs}} \mid \theta)$, not a distribution over $\theta$).



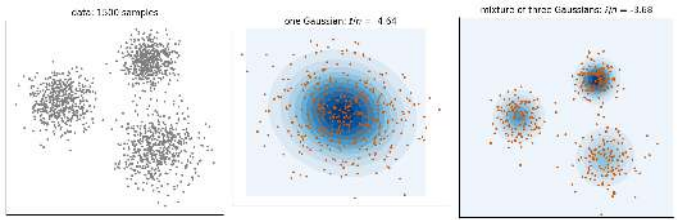

A coin tossed 10 times, $p(k \mid \theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$. Left: $p(7 \mid 0.5) = 0.117$. Right: $k = 7$ observed, $\arg\max_\theta L = 0.7$, $\int_0^1 L\, d\theta = 1/11$.

# Maximum Likelihood

Independent samples $\mathbf{x}_1, \ldots, \mathbf{x}_n$, a model $p_\theta(\mathbf{x})$:

$$\ell(\theta) = \log \prod_{i=1}^{n} p_\theta(\mathbf{x}_i) = \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i), \qquad \theta^* = \arg\max_\theta \ell(\theta)$$

$\frac{1}{n}\ell \to -H(p) - \mathrm{KL}(p \,\|\, p_\theta)$ for $n \to \infty$: maximum likelihood minimises $\mathrm{KL}(p \,\|\, p_\theta)$.




$p_\theta = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$, $\pi = \mathrm{softmax}(\mathbf{a})$, $\Sigma_k = L_k L_k^\top$; gradient ascent on $\ell$; orange: 400 new samples. $\ell/n = -4.64$ ($K = 1$), $-3.68$ ($K = 3$). For images: a neural network.

# 2

# Variational Autoencoders



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# A Latent-Variable Model

A fixed **prior**, and a network giving pixel probabilities $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$:

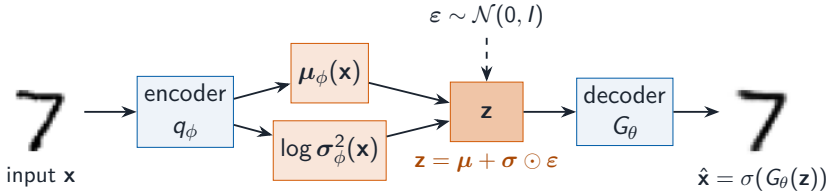
$$\mathbf{z} \sim \mathcal{N}(0, I_d), \qquad p_\theta(\mathbf{x} \mid \mathbf{z}) = \prod_{j=1}^{784} \hat{x}_j^{x_j} (1 - \hat{x}_j)^{1 - x_j}, \qquad p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$

$-\log p_\theta(\mathbf{x}|\mathbf{z}) = -\sum_j [x_j \log \hat{x}_j + (1 - x_j) \log(1 - \hat{x}_j)] = \mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}})$ (binary cross-entropy).

- **generating is easy:** draw $\mathbf{z} \sim \mathcal{N}(0, I)$, decode
- **maximum likelihood is hard:** the integral has no closed form, and sampling it from the prior fails: almost no $\mathbf{z}$ produces a given $\mathbf{x}$. Trained $d = 16$ VAE, $K = 1000$ samples, test average: $\log p_\theta(\mathbf{x}) \approx -154.8$ (prior), $-93.3$ (importance sampling with the encoder)
- the fix: an **encoder** $q_\phi(\mathbf{z} \mid \mathbf{x})$ proposes the $\mathbf{z}$ that could have produced $\mathbf{x}$

D. P. Kingma, M. Welling, ICLR 2014, arXiv:1312.6114; D. J. Rezende, S. Mohamed, D. Wierstra, ICML 2014, PMLR 32, 1278

# Variational Autoencoder: the Architecture




- encoder $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}^2_\phi(\mathbf{x}))$: a Gaussian *cloud* instead of a point; outputs $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$
- training decodes a random point of the cloud, so every $\mathbf{z}$ near $\boldsymbol{\mu}(\mathbf{x})$ must decode to something like $\mathbf{x}$
- generation skips the encoder: $\mathbf{z} \sim \mathcal{N}(0, I)$, $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$
- MNIST: $784 \to 400 \to 200 \to (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$ and $d \to 200 \to 400 \to 784$, ReLU, $d = 2, 16$

# The Evidence Lower Bound

**Goal:** maximise $\log p_\theta(\mathbf{x}) = \log \int p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$, an integral we cannot compute.

**1.** Bayes' rule, for every $\mathbf{z}$: $\log p_\theta(\mathbf{x}) = \log p_\theta(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}) - \log p_\theta(\mathbf{z}|\mathbf{x})$.

**2.** Insert the encoder $q_\phi(\mathbf{z}|\mathbf{x})$ and average over it, $\mathbb{E}_q[f] = \int q(\mathbf{z}) f(\mathbf{z})\, d\mathbf{z}$; $\mathrm{KL}(q\|p) = \mathbb{E}_q \log(q/p)$:

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}\,|\,\mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\mathrm{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{\mathrm{KL}\big(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p_\theta(\mathbf{z}|\mathbf{x})\big)}_{\geq 0}$$

**3.** $\mathrm{KL} \geq 0$ (Jensen: $-\mathrm{KL} = \mathbb{E}_q \log \frac{p}{q} \leq \log \mathbb{E}_q \frac{p}{q} = \log 1 = 0$) $\Rightarrow \log p_\theta(\mathbf{x}) \geq \mathrm{ELBO}$: the **E**vidence **L**ower **BO**und, tight when $q_\phi$ is the true posterior.
Maximising the ELBO raises $\log p_\theta(\mathbf{x})$ and pulls $q_\phi$ towards $p_\theta(\mathbf{z}|\mathbf{x})$. All of it is computable; the KL of two Gaussians:

$$\mathrm{KL}\big(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \,\|\, \mathcal{N}(0, I)\big) = \frac{1}{2} \sum_{k=1}^{d} \left(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1\right)$$

# The VAE Loss, and a Variational Free Energy

Loss per image, $-\mathrm{ELBO}$, with one sample $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$:

$$\mathcal{L}(\mathbf{x}) = \underbrace{\mathrm{BCE}(\mathbf{x}, \sigma(G_\theta(\mathbf{z})))}_{\text{reconstruction}} + \underbrace{\frac{1}{2}\sum_k \left(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1\right)}_{\text{stay close to the prior}}$$

- reconstruction: samples of each cloud decode back to $\mathbf{x}$; alone, it shrinks the clouds to points (an autoencoder)
- KL: pulls every cloud to $\mathcal{N}(0, I)$; the clouds overlap and fill the prior, so a fresh $\mathbf{z} \sim \mathcal{N}(0, I)$ decodes to a valid image

**Free energy.** States $\mathbf{z}$, energy $E(\mathbf{z})$, temperature $T$: Boltzmann $e^{-E/T}/Z$, $F = -T \log Z$; for a trial distribution $q$, $F[q] = \langle E \rangle_q - TS[q]$, $S[q] = -\mathbb{E}_q \log q$. With $E(\mathbf{z}) = -\log[p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})]$, $T = 1$: $e^{-E}/Z = p_\theta(\mathbf{z}|\mathbf{x})$, $Z = p_\theta(\mathbf{x})$, and

$$F[q] = F + \mathrm{KL}(q \,\|\, e^{-E}/Z) \geq F \quad \text{(Gibbs–Bogoliubov)}, \qquad F[q] = -\mathrm{ELBO}$$

# The Reparametrisation Trick

$\mathbb{E}_q$ is estimated with one sample of $\mathbf{z}$ per image. But "draw $\mathbf{z}$ from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$" has no derivative with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$.

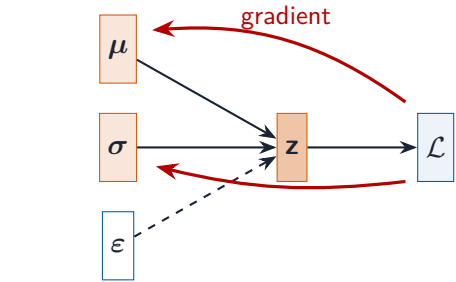
Move the randomness out of the network:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, I), \qquad \boldsymbol{\sigma} = e^{\frac{1}{2}\log \boldsymbol{\sigma}^2}$$

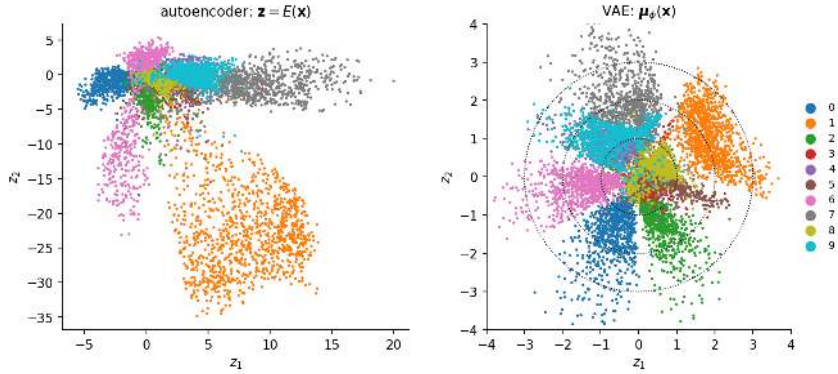
$$\frac{\partial z_k}{\partial \mu_k} = 1, \qquad \frac{\partial z_k}{\partial \sigma_k} = \varepsilon_k$$

$\boldsymbol{\varepsilon}$ does not depend on the parameters; backpropagation runs from the loss through decoder, $\mathbf{z}$ and encoder.

In JAX, $\boldsymbol{\varepsilon}$ comes from an explicit key: `eps = jax.random.normal(key, mu.shape)`.

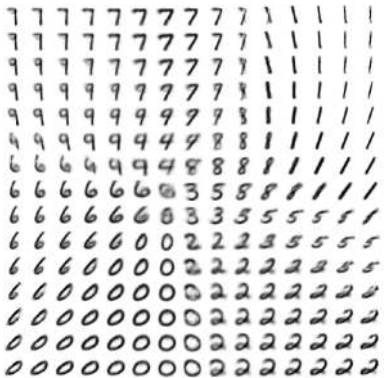


# The Latent Plane: Autoencoder and VAE

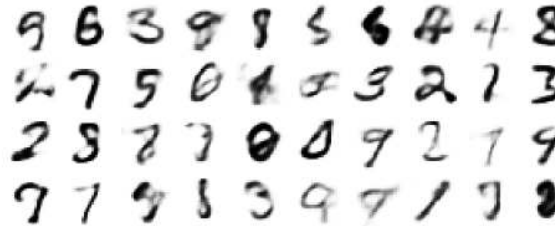



$d = 2$, the 10 000 test digits. Left: $\mathbf{z} = E(\mathbf{x})$; right: $\boldsymbol{\mu}_\phi(\mathbf{x})$, dotted: $|\mathbf{z}| = 1, 2, 3$. Standard deviation of the coordinates: 3.8, 7.5 (AE), 1.03, 1.09 (VAE). The VAE codes fill the region where $\mathcal{N}(0, I)$ puts its probability: **we know where to sample.**

# New Digits from $\mathbf{z} \sim \mathcal{N}(0, I)$



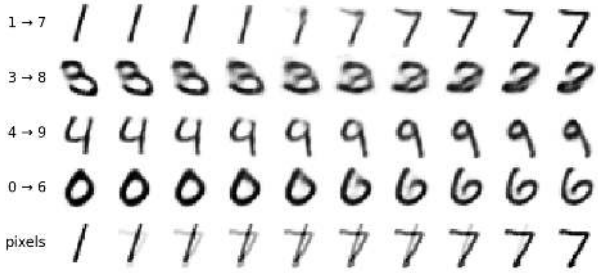
$d = 2$: $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z}$ on a grid in $[-2.5, 2.5]^2$



$d = 16$: 40 samples, $\mathbf{z} \sim \mathcal{N}(0, I)$

- every grid point decodes to a digit
- not copies: $\langle \min_i \|\hat{\mathbf{x}} - \mathbf{x}_i\| \rangle$ over the training set, 4.48 (generated), 4.19 (real test)
- blurry: we show the pixel probabilities $\hat{x}_j$

# Interpolation in the Latent Space

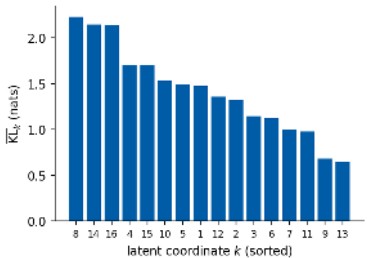



$\mathbf{z}(t) = (1-t)\,\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\,\boldsymbol{\mu}_\phi(\mathbf{x}_b)$, $t \in [0,1]$, decoded by the $d = 16$ VAE; last row: $(1-t)\,\mathbf{x}_a + t\,\mathbf{x}_b$.

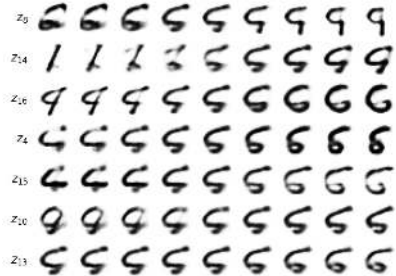
- latent line: plausible digits throughout; the 1 bends into a 7, the 4 closes into a 9
- pixel line: two superposed digits

**Straight lines in the latent space stay among the data.**

# Meaning of the Latent Coordinates




$$\overline{\mathrm{KL}_k} = \left\langle \tfrac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1) \right\rangle_{\mathbf{x}}$$




$\boldsymbol{\mu}_\phi(\mathbf{x})$ of one test 5, one $z_k$ from $-3$ to 3; top 6 by $\overline{\mathrm{KL}_k}$, and the smallest

- all 16 coordinates are used, unequally: $\overline{\mathrm{KL}_k}$ from 0.65 to 2.23 nats ($\approx 0$: ignored)
- one coordinate changes slant, width, thickness, and often the digit itself

# Limits of Interpreting the Latent Space

## We can read

- neighbourhoods: nearby $\mathbf{z}$, similar images
- straight paths: continuous deformations
- usage of each coordinate: $\overline{\mathrm{KL}}_k$
- where the data are: $\mathcal{N}(0, I)$

## We cannot read

- the meaning of one axis: $\mathcal{N}(0, I)$ and the KL are unchanged by a rotation $\mathbf{z} \to R\mathbf{z}$, and the decoder absorbs it, $G'(\mathbf{z}) = G(R^\top \mathbf{z})$
- one property per axis ("disentanglement"): not guaranteed without labels or further assumptions

**Telling the model what to draw, with a label: the conditional VAE.**

F. Locatello et al., *Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations*, ICML 2019, PMLR 97, 4114

# 3

# Judging Synthetic Data



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# A Classifier as a Referee

Pictures are judged by eye. To *measure* what a generator produces, use a referee trained on the **real** data, with known accuracy on real test data, and apply it unchanged to synthetic images.

- CNN: $3 \times 3$ convolutions with 16 and 32 channels, ReLU, each followed by $2 \times 2$ max-pooling ($28 \to 14 \to 7$); dense 128; 10 outputs $f_c(\mathbf{x})$
- trained with the cross-entropy, the negative log-likelihood of the labels (Adam, 4 epochs):

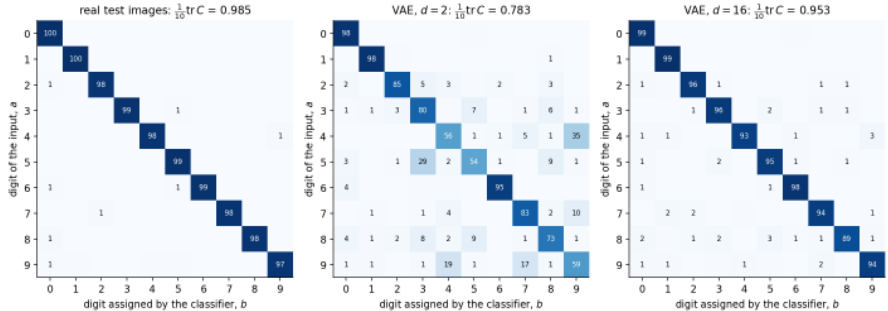
$$\mathcal{L} = -\frac{1}{n}\sum_i \log p(y_i \,|\, \mathbf{x}_i), \qquad p(c \,|\, \mathbf{x}) = \frac{e^{f_c(\mathbf{x})}}{\sum_{c'} e^{f_{c'}(\mathbf{x})}}$$

test accuracy on 10 000 real digits

**0.9854**

**Every test of synthetic data is first run on real data: the baseline.**

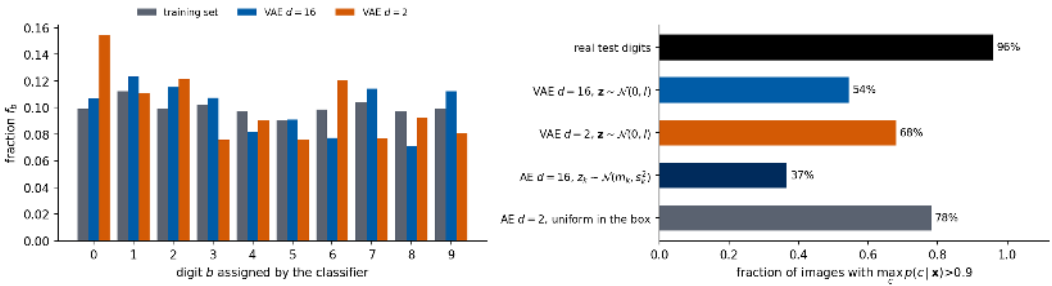
# Confusion Matrices of VAE Reconstructions




$C_{ab} = \#\{\text{inputs of digit } a \text{ assigned to } b\}/\#\{\text{inputs of digit } a\}$; row $a$: digit of the real test input $\mathbf{x}$; column $b$: classifier label of $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z} \sim q_\phi(\mathbf{z}\,|\,\mathbf{x})$; in %.

- $d = 16$: $\frac{1}{10}\mathrm{tr}\,C = 0.953$ against 0.985 for real images: the digit survives the latent space
- $d = 2$: look-alike digits share a region of the plane; worst: $4 \to 9$: 35%

# Digit Distribution of the Generated Samples




Left: $f_b = \#\{\text{samples assigned to } b\}/N$, $N = 10\,000$, $\mathbf{z} \sim \mathcal{N}(0, I)$. Right: fraction with $\max_c p(c \mid \mathbf{x}) > 0.9$.

- $d = 16$: every digit, near its training frequency; $d = 2$: too many 0s, too few 3s, 5s, 7s
- many VAE samples are blurry or between two digits; the $d = 2$ AE decodes digit-like images in most of its box, but a uniform box is no model of the data

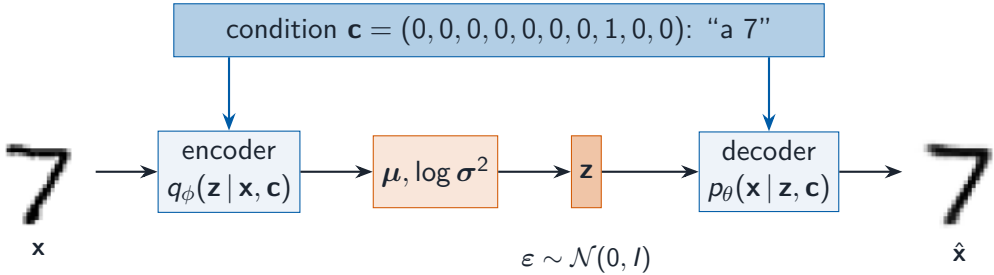
# 4

# Conditional Generation



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# Conditional VAE: the Architecture




- $\mathbf{c}$: a label known for every training image, here the **one-hot** digit, $c_j = \delta_{j,y}$; concatenated to the inputs of both networks: $[\mathbf{x}, \mathbf{c}] \in \mathbb{R}^{794}$ and $[\mathbf{z}, \mathbf{c}] \in \mathbb{R}^{d+10}$
- prior $p(\mathbf{z}) = \mathcal{N}(0, I)$ for every $\mathbf{c}$; loss: BCE + KL as before; one-hot, since a single number $y$ would put 3 "between" 2 and 4

K. Sohn, H. Lee, X. Yan, *Learning Structured Output Representation using Deep Conditional Generative Models*, NeurIPS 2015

# The Conditioned ELBO and Generation on Demand

The same derivation with $\mathbf{c}$ carried along gives a lower bound on the conditional likelihood:

$$\log p_\theta(\mathbf{x} \mid \mathbf{c}) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}\mid\mathbf{x},\mathbf{c})}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{c})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{c}) \,\big\|\, \mathcal{N}(0, I)\big)$$

**Training**

pairs $(\mathbf{x}_i, \mathbf{c}_i)$; $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}) + \boldsymbol{\sigma}_\phi(\mathbf{x}, \mathbf{c}) \odot \varepsilon$; loss $\mathrm{BCE}(\mathbf{x}, \sigma(G_\theta(\mathbf{z}, \mathbf{c}))) + \mathrm{KL}$

**Generation**

choose $\mathbf{c}$, draw $\mathbf{z} \sim \mathcal{N}(0, I)$, compute $\sigma(G_\theta(\mathbf{z}, \mathbf{c}))$; no encoder needed

- $\mathbf{c}$ already tells the decoder *which* digit; the KL term penalises every nat stored in $\mathbf{z}$, so $\mathbf{z}$ keeps only what $\mathbf{c}$ does not say: the **style** (slant, width, size)
- $\mathbf{c}$ can be any vector: a one-hot class, a temperature $T$, a point $(g, T)$ of a phase diagram; MNIST: $d = 8$

# Digits on Demand



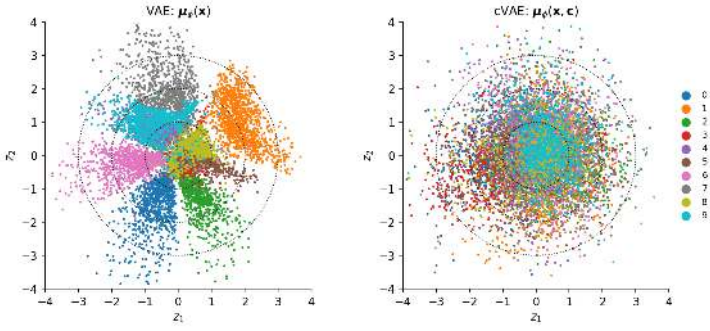

$\sigma(G_\theta(\mathbf{z}, \mathbf{c}))$; row: $\mathbf{c}$ = digit 0 to 9; column: one $\mathbf{z} \sim \mathcal{N}(0, I)$, the same in every row.

Along a column the digit changes and the style stays: $\mathbf{z}$ carries the style, $\mathbf{c}$ the identity.



Style transfer: a real test digit $\mathbf{x}$ (left), $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}_y)$, decoded with $\mathbf{c} = 0, \ldots, 9$.

# The Latent Plane of a cVAE Holds Only the Style

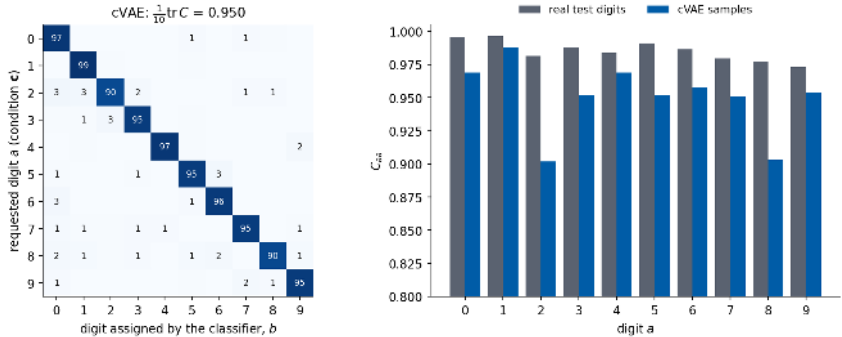





cVAE, $\mathbf{c} = 7$, $\mathbf{z}$ on a grid over $[-2, 2]^2$

$d = 2$; test digits at $\boldsymbol{\mu}_\phi(\mathbf{x})$ (VAE) and $\boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c})$ (cVAE). Digit predicted from the nearest class mean in the latent plane: 0.73 (VAE), 0.18 (cVAE), chance 0.1. In the cVAE all digits share one disc: $\mathbf{z}$ is a map of style.

# The Requested Digit Is Generated




1000 samples per requested digit, $\mathbf{z} \sim \mathcal{N}(0, I)$.
$C_{ab} = \#\{\text{samples generated with } \mathbf{c} = a \text{ assigned to } b\}/1000$; row $a$: requested digit; column $b$: classifier label; in %. Right: the diagonal $C_{aa}$, next to the classifier on real test digits. Mean $C_{aa}$: 0.950; largest confusion $2 \to 0$: 3%.

# From Digits to Spins

The protocol of this part, ready for physics:

1. a generator conditioned on what we want: $\mathbf{c}$ = the digit $\longrightarrow$ $\mathbf{c} = T$ for the 2D Ising model, $\mathbf{c} = (g, T)$ for the $J_1$–$J_2$ model
2. referees trained on **real** data, checked on real test data first $\longrightarrow$ networks trained on Monte Carlo configurations
3. the same tests on synthetic samples

**Physics has a stricter referee**

the true distribution is known, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$: energy, magnetisation, fluctuations and correlations of generated configurations can be compared with Monte Carlo at every $T$

**The question**

do cVAE samples obey the same physics as the Monte Carlo samples they were trained on, including near the phase transition?

# 5

# Generated Spins and the Hamiltonian



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# The Hamiltonian Fixes the Distribution

spins $s_i = \pm 1$ on an $L \times L$ lattice, periodic boundaries, $N = L^2$, $k_B = J = J_1 = 1$:

$$H_{\text{Ising}} = -\sum_{\langle ij \rangle} s_i s_j, \qquad H_{J_1 J_2} = -\sum_{\langle ij \rangle} s_i s_j + g \sum_{\langle\langle ij \rangle\rangle} s_i s_j, \qquad p(\mathbf{s} \mid T) = \frac{e^{-H(\mathbf{s})/T}}{Z}$$

| observable | measures |
|---|---|
| $e = H/N$ | energy per spin: satisfied and broken bonds (local structure) |
| $m = \frac{1}{N}\sum_i s_i$, $m_{\text{str}} = \left(m^2_{(\pi,0)} + m^2_{(0,\pi)}\right)^{1/2}$ | ferromagnetic order, stripe order ($J_1$–$J_2$, $g > 1/2$) |
| $c = N\,\mathrm{Var}(e)/T^2$, $\chi = N\,\mathrm{Var}(\lvert m\rvert)/T$ | energy and magnetisation fluctuations; peak at a transition |
| $U_4 = 1 - \langle m^4\rangle / 3\langle m^2\rangle^2$ | shape of $P(m)$: 2/3 ordered, $\to 0$ disordered |
| $G(r) = \langle s_{\mathbf{x}} s_{\mathbf{x}+r\hat{\mathbf{e}}}\rangle$ | how far spin correlations extend |
| $P(e)$, $P(m)$ | full distributions, not only averages |

2D Ising: $T_c = 2/\ln(1+\sqrt{2}) = 2.269$ (L. Onsager, Phys. Rev. **65**, 117 (1944); C. N. Yang, Phys. Rev. **85**, 808 (1952)). A generator of spin configurations has to reproduce $p(\mathbf{s} \mid T)$, not only its look.

# The cVAE for Spins: a Product of Bernoulli Factors

condition $c = T$ (Ising) or $c = (g, T)$ ($J_1$–$J_2$); bits $b_i = (s_i + 1)/2$; one logit $\ell_i(\mathbf{z}, c)$ per site

$$\log p_\theta(\mathbf{s} \mid c) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{s},c)}\Big[\sum_{i=1}^{N} b_i \log\sigma(\ell_i) + (1-b_i)\log\big(1-\sigma(\ell_i)\big)\Big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{s}, c) \,\big\|\, p(\mathbf{z} \mid c)\big)$$

prior $p(\mathbf{z}|c) = \mathcal{N}(0, I)$ as in the MNIST cVAE, or learned (below)

## Given $\mathbf{z}$, the spins are independent

$p_\theta(\mathbf{s} \mid \mathbf{z}, c) = \prod_i p(s_i \mid \mathbf{z}, c)$: a mean-field-like product, $\langle s_i \rangle_{\mathbf{z}} = \tanh(\ell_i/2)$,

$$\mathrm{Cov}(s_i, s_j) = \mathrm{Cov}_{\mathbf{z}}\big(\tanh\tfrac{\ell_i}{2}, \tanh\tfrac{\ell_j}{2}\big), \quad i \neq j$$

all correlations must pass through the $d = 16$ latent variables

## Two ways to generate

$\mathbf{z} \sim p(\mathbf{z} \mid c)$, then

- **Bernoulli**: $s_i = +1$ with probability $\sigma(\ell_i)$
- **most likely**: $\hat{s}_i = \operatorname{sign} \ell_i$

$H$ is a sum of neighbour products $s_i s_j$: independent draws break bonds at random. $e$, $c$, $G(1)$ are the sharp tests.

D. P. Kingma, M. Welling, arXiv:1312.6114; K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

# Two Kinds of Tests, Each with a Real-vs-Real Baseline

2D Ising, $L = 30$, 37 temperatures $T \in [1.5, 3.5]$, 256 Swendsen–Wang snapshots per $T$, split into **training** (128), **held-out** $A$ (64), **held-out** $B$ (64); synthetic: 64 snapshots per $T$

## 1. Networks trained on Monte Carlo data

CNN regressor $\hat{T}(\mathbf{s})$, loss $(\hat{T} - T)^2$;
CNN phase classifier trained at $T \leq 1.8$ and $T \geq 2.8$, confidence $\max(p_0, p_1)$

## 2. The Hamiltonian

$\langle e \rangle$, $\langle |m| \rangle$, $c$, $\chi$, $U_4$, $G(r)$ and the distributions $P(e)$, $P(m)$, compared by $W_1 = \frac{1}{n} \sum_k |x_{(k)} - y_{(k)}|$

## 3. The generator itself

$-$ELBO on training and held-out data; active latent units ($\mathrm{KL}_j > 0.01$); classifier two-sample test (C2ST): accuracy of a CNN separating real from synthetic, 0.5 = indistinguishable; memorisation: $d_{\mathrm{NN}} = \min_{\mathbf{t} \in \mathrm{train}}(1 - |\mathbf{s} \cdot \mathbf{t}|/N)/2$

## Noise units

$R_O = \mathrm{mean}_T |O_{\mathrm{syn}} - O_A| \,/\, \mathrm{mean}_T |O_B - O_A|$;
$R \approx 1$: as close to $A$ as an independent Monte Carlo set

every test runs first on held-out Monte Carlo data, then identically on the synthetic data at the same $T$ C2ST: D. Lopez-Paz, M. Oquab, arXiv:1610.06545

# Three cVAEs, Each Adding One Piece of Physics

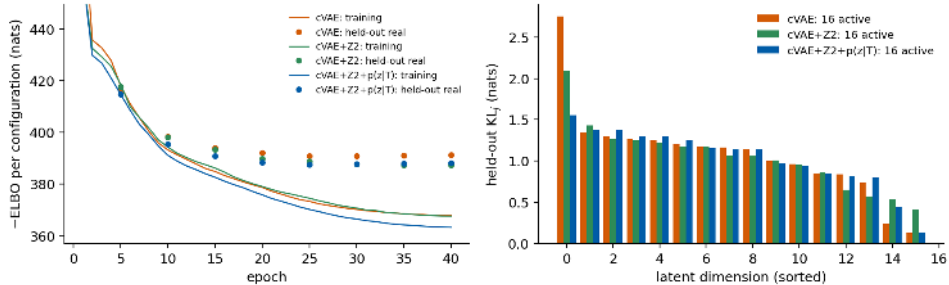
dense encoder $N{+}1 \to 512 \to 256 \to (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$, latent dimension $d = 16$; dense decoder $d{+}1 \to 256 \to 512 \to N$ logits; 40 epochs, Adam

- **cVAE**: prior $\mathcal{N}(0, I)$, raw data (the MNIST model)
- **cVAE+Z2**: the symmetry $H(-\mathbf{s}) = H(\mathbf{s})$ imposed by construction: train on the $m \geq 0$ representative of every snapshot, flip each generated snapshot with probability $\frac{1}{2}$
- **cVAE+Z2+p(z|T)**: conditional prior $p_\psi(\mathbf{z} \mid T) = \mathcal{N}(\boldsymbol{\mu}_\psi(T), \operatorname{diag} \boldsymbol{\sigma}^2_\psi(T))$ from a small network $T \to 64 \to (\boldsymbol{\mu}_\psi, \log \boldsymbol{\sigma}^2_\psi)$: a snapshot reveals its $T$, so the encoded snapshots of different $T$ occupy different regions of $\mathbf{z}$-space; $\mathbf{z} \sim \mathcal{N}(0, I)$ draws from all of them, $\mathbf{z} \sim p_\psi(\mathbf{z}|T)$ from the region of $T$

$$\mathrm{KL}(q \,\|\, p_\psi) = \frac{1}{2} \sum_{j=1}^{d} \Big[ \log \frac{\sigma^2_{\psi,j}}{\sigma^2_j} + \frac{\sigma^2_j + (\mu_j - \mu_{\psi,j})^2}{\sigma^2_{\psi,j}} - 1 \Big]$$

prior network: K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

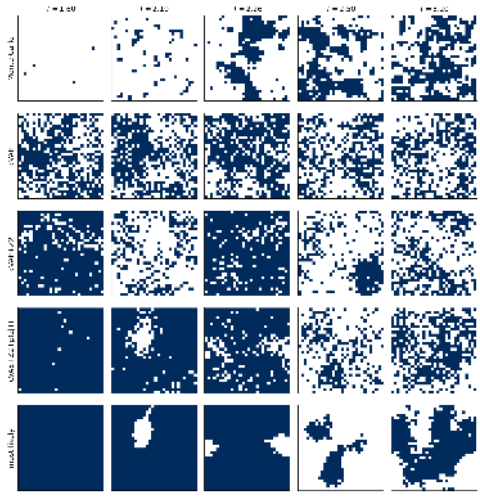
# Training Health Does Not Rank the Three Models




left: $-$ELBO per configuration (nats; uniform spins $N \ln 2 = 624$), training and held-out real data; right: held-out $\mathrm{KL}_j$ per latent dimension

| | cVAE | cVAE+Z2 | cVAE+Z2+p(z\|T) |
|---|---|---|---|
| $-$ELBO training / held-out | 368 / 391 | 368 / 387 | 363 / 388 |
| active units ($\mathrm{KL}_j > 0.01$) of 16 | 16 | 16 | 16 |
| spins reproduced from $\mathbf{z} = \boldsymbol{\mu}$ | 80 % | 81 % | 81 % |

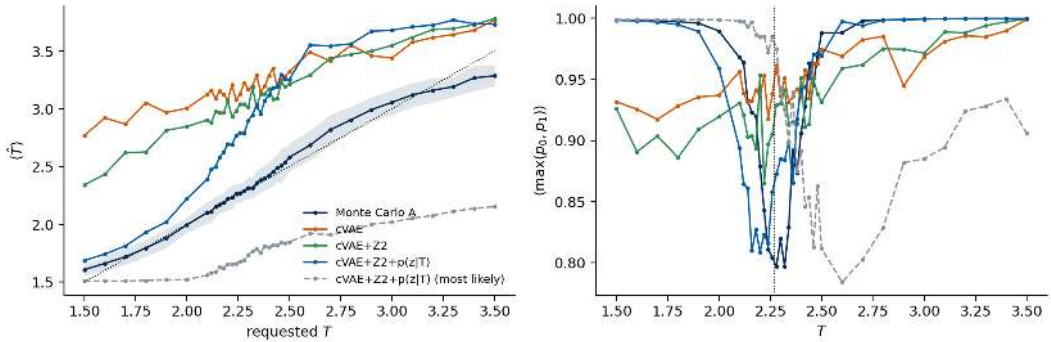
# Monte Carlo and Synthetic Ising Configurations



rows: Monte Carlo; cVAE; cVAE+Z2; cVAE+Z2+p(z|T) with Bernoulli sampling; the same $\mathbf{z}$ with $\hat{s}_i = \operatorname{sign} \ell_i$

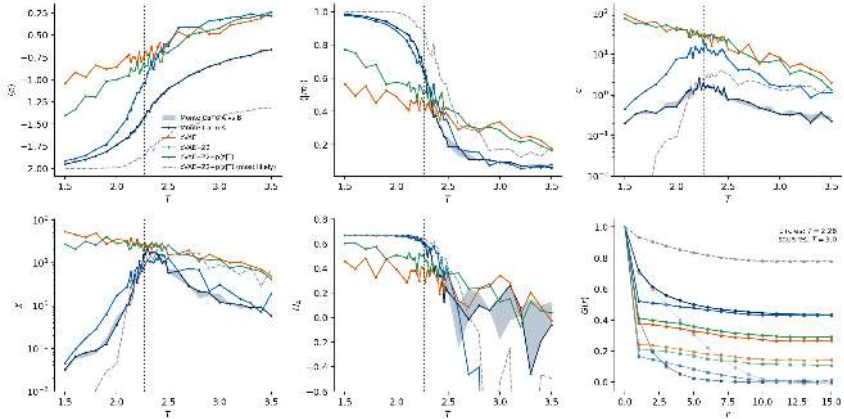
- cVAE and +Z2: the requested $T$ is barely visible
- +p(z|T): ordered below $T_c$, disordered above; salt-and-pepper noise inside domains
- most likely configuration: no noise, but large smooth domains even at $T = 3.2$

# Networks Trained on Monte Carlo Data Judge the Samples



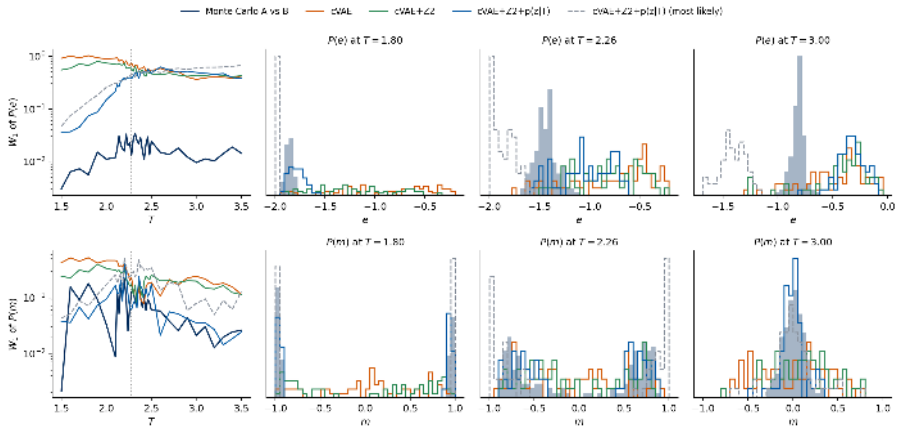

left: $\langle \hat{T} \rangle$ vs requested $T$ (band: $\pm$ std on $A$); right: $\langle \max(p_0, p_1) \rangle_T$. RMSE of $\hat{T}$: Monte Carlo $A$ 0.12, $B$ 0.12; cVAE 0.98; +Z2 0.84; +p(z|T) 0.62; most likely 0.71. Below $T_c$ only +p(z|T) is read at the requested temperature; above $T_c$ every Bernoulli model looks too hot.

# The Order Parameter Is Reproduced, the Energy Is Not



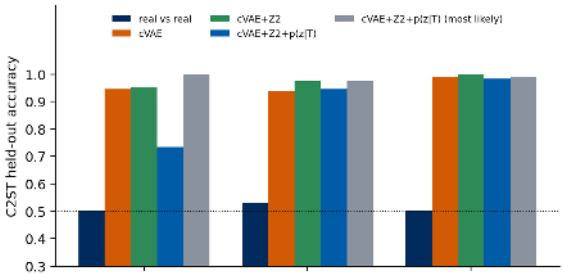
$\langle e \rangle$, $\langle |m| \rangle$, $c$, $\chi$, $U_4$ vs $T$; $G(r)$ at $T = 2.26$ (circles) and $T = 3.0$ (squares); band: Monte Carlo $A$ vs $B$. Noise ratios of +p(z|T): $R_{|m|} = 2.8$, $R_\chi = 2.0$, $R_{W_1(m)} = 1.3$; $R_e = 34.8$, $R_c = 39.0$, $R_{W_1(e)} = 21.6$. At $T = 2.26$: $\langle e \rangle = -1.43$ (Monte Carlo), -1.04 (Bernoulli), -1.86 (most likely); $G(1) = -\langle e \rangle / 2$: the energy is the nearest-neighbour correlation

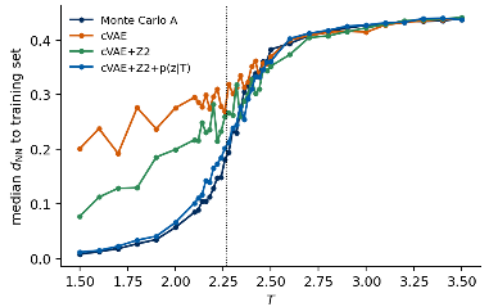
# Distributions of Energy and Magnetisation




$W_1$ between synthetic and Monte Carlo $A$ vs $T$, against $W_1(A, B)$; $P(e)$, $P(m)$ at $T = 1.8, 2.26, 3.0$. Fraction of snapshots below $T_c$ with $|m| < \langle|m|\rangle_{\mathrm{MC}}/2$: Monte Carlo 1 %, cVAE 45 %, +Z2 27 %, +p(z|T) 1 %. $P(m)$ is right, $P(e)$ is shifted and too broad.

# A Classifier Separates Real from Synthetic; Nothing Is Copied

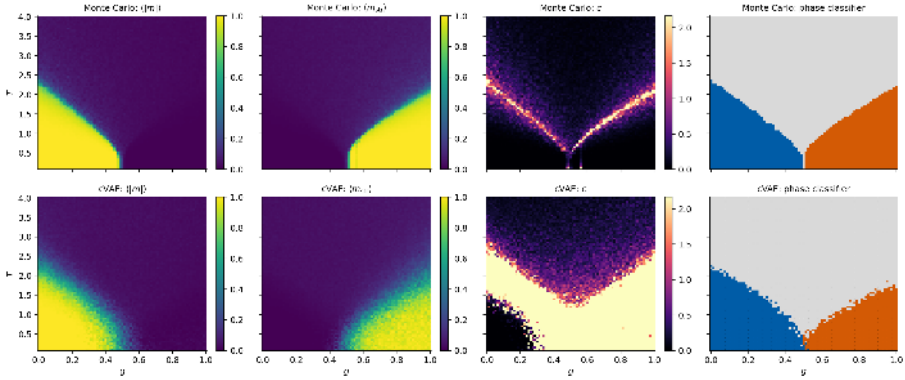



C2ST held-out accuracy at $T = 1.8/2.26/3.0$: real vs real 0.50 / 0.53 / 0.50; +p(z|T) 0.73 / 0.95 / 0.98; most likely 1.00 / 0.98 / 0.99

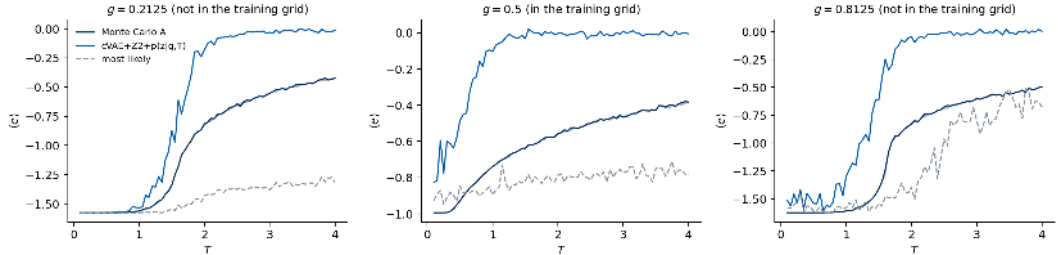
median $d_{\mathrm{NN}}$ to the training set at the same $T$: Monte Carlo 0.03 / 0.18 / 0.43; +p(z|T) 0.03 / 0.20 / 0.43: new configurations, no memorisation

# $J_1$–$J_2$: the Phase Diagram Is Interpolated




$81 \times 79$ grid in $(g, T)$, 100 snapshots per point; cVAE+Z2+p(z|g,T) trained on every second $g$ and every second $T$ (dots), generated everywhere; columns: $\langle |m| \rangle$, $\langle m_{\mathrm{str}} \rangle$, $c$, majority phase of a CNN classifier trained on real data (FM, stripes, PM). Same phase as Monte Carlo $A$: $B$ 99.7 %; synthetic 95.3 % at seen, 95.4 % at unseen $(g, T)$; most likely configurations 35.4 %.

# $J_1$–$J_2$: Order Parameters Right, Energy Wrong




| $R$ (noise units) | $\langle \lvert m \rvert \rangle$ | $\langle m_{\mathrm{str}} \rangle$ | $\langle e \rangle$ | $c$ |
|---|---|---|---|---|
| seen $(g, T)$ | 4.4 | 6.4 | 64.0 | 2279 |
| unseen $(g, T)$ | 4.8 | 5.9 | 63.9 | 1517 |

−ELBO held-out: 476 (seen), 482 (unseen); stripes alternating along $x$: 0.50 (Monte Carlo), 0.48 (cVAE). Paramagnet: synthetic $\langle e \rangle \approx 0$, the value for independent spins

# What the Generator Reproduces and What It Does Not

## Reproduced

- Ising $\langle|m|\rangle$, $\chi$, $P(m)$: $R = 2.8$, 2.0, 1.3; $J_1$–$J_2$ $\langle|m|\rangle$, $\langle m_{\mathrm{str}}\rangle$: $R \approx 4.8$, 5.9 (unseen)
- the temperature seen by a regressor below $T_c$, the phase seen by classifiers trained on real data
- the $J_1$–$J_2$ phase diagram at unseen $(g, T)$; both stripe orientations
- new configurations, not copies of the training set

needed: $\mathbb{Z}_2$ symmetry of $H$ by construction, and a conditional prior $p(\mathbf{z}|c)$

## Not reproduced

- $\langle e\rangle = H/N$: $R = 34.8$ (Ising), 63.9 ($J_1$–$J_2$)
- $c$, $P(e)$: energy fluctuations an order of magnitude too large
- a C2ST separates real from synthetic at every $T$

given $\mathbf{z}$ the decoder is a product over sites, a mean-field-like distribution: nearest-neighbour correlations beyond what 16 latent variables carry are lost. Bernoulli sampling breaks bonds, $\operatorname{sign}\ell_i$ over-smooths.

**Looking right is not obeying $H$:** synthetic configurations answer questions about the phase; they are not samples of $e^{-H/T}/Z$ and do not replace Monte Carlo for thermodynamics.

# Summary

- an **autoencoder** learns a latent space, not a distribution on it: random latent vectors decode to non-digits
- **maximum likelihood** fits $p_\theta$ to data, i.e. minimises $\mathrm{KL}(p \,\|\, p_\theta)$
- a **VAE** maximises $\mathrm{ELBO} = \mathbb{E}_q \log p_\theta(\mathbf{x}|\mathbf{z}) - \mathrm{KL}(q \,\|\, p)$, trainable through the reparametrisation trick; the KL term shapes the latent space into $\mathcal{N}(0, I)$: sampling, interpolation and traversals work, single axes need not have a meaning
- a **classifier trained on real data** referees synthetic data: confusion matrices, class distributions, confidence, always against the real-data baseline
- a **conditional VAE** adds a known label $\mathbf{c}$ to encoder and decoder: $\mathbf{c}$ selects what is generated, $\mathbf{z}$ carries the rest
- **spins**: a cVAE conditioned on $T$ or $(g, T)$, with the $\mathbb{Z}_2$ symmetry and a conditional prior, reproduces the phase, the order parameter and the $J_1$–$J_2$ phase diagram at unseen $(g, T)$; it misses the energy and its fluctuations: a product of independent spins given $\mathbf{z}$ is not a sample of $e^{-H/T}/Z$

# Tutorial: Hands-on

## Hands-on (Lab 024)

- train the $d = 16$ VAE without the KL term and sample from $\mathcal{N}(0, I)$; then multiply the KL term by $\beta = 4$: how many active coordinates remain, and how sharp are the samples?
- give the cVAE a mixed condition $\mathbf{c} = \frac{1}{2}(\mathbf{e}_4 + \mathbf{e}_9)$: what does it draw, and what does the classifier say?
- replace the one-hot condition by the single number $y/9$ and measure the confusion matrix again

# Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar

